Anshu Rajendra Borkar

ms23btech11006@iith.ac.in | anshuborkar24@gmail.com | +91 8329080611 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.04
SNBP	Class XII (CBSE)	2022	88.00%
Kendriya Vidyalay	Class X (CBSE)	2020	94.20%

ACHIEVEMENTS & POSITIONS OF RESPONSIBILITY

- Assistant Manager in Ecell .
- Inter IIT Chess 2024
- NSO Chess
- Vibes Member

SKILLS

- **Programming Languages:** Python, C++
- **Core CS Concepts:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOPs)
- **AI & ML Domains:** Deep Learning, Machine Learning, LLMs, Natural Language Processing (NLP), Computer Vision
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, LangChain, OpenCV, Scikit-learn, NumPy, Pandas, Matplotlib
- **Tools & Cloud:** Git, GitHub, AWS

RELEVANT COURSES

- ID1063 - Introduction to Programming
- ID1050 - Artificial Intelligence

PROJECTS

FULL STACK NLP FOOD DELIVERY CHATBOT

Engineered an end-to-end conversational AI chatbot for automated food ordering and order tracking using Dialogflow, FastAPI, and MySQL.

- **Engineered** a conversational NLU engine in **Dialogflow** with **5+ custom intents** and entity recognition for **9 food items**, enabling automated multi-turn ordering.
- **Developed** a high-performance **FastAPI** backend webhook server with in-memory session cart management and **Regex session extraction**, ensuring **100% data fidelity**.
- **Architected** a robust **MySQL** database schema featuring stored procedures and user-defined functions for real-time order persistence and automated bill calculation.

SELF-TRAINING ANTICHESS AI ENGINE

Unsupervised deep reinforcement learning chess engine combining Monte Carlo Tree Search (MCTS) and Residual Neural Networks in PyTorch.

- **Engineered** a self-training Antichess AI model using a **4-block ResNet architecture** and **Monte Carlo Tree Search (MCTS)** in **PyTorch**, evaluating over **14,800 board positions** via self-play.
- **Optimized** the neural network pipeline by implementing **Stochastic Gradient Descent (SGD)** and custom **tuple data unpacking**, eliminating array shape mismatches and boosting training execution speed by **2.5x** over baseline frameworks.
- **Developed** a real-time **800x800 interactive desktop GUI** with forced-capture rule validation using **Pygame** and **python-chess**, integrating a **zero-temperature MCTS search mode** for competitive human-vs-AI gameplay.

HYBRID LOG CLASSIFICATION SYSTEM

Engineered an enterprise log monitoring backend to automatically categorize raw software log lines into functional domains.

- **Architected** a 3-tier hybrid classification pipeline using **Python RegEx**, **BERT (MiniLM) embeddings** with **Logistic Regression**, and **Llama 3.3 / DeepSeek R1 LLMs**, reducing overall inference latency and API token expenses.
- **Extracted** recurring log patterns by clustering sentence transformer vectors via **DBSCAN**, enabling automated RegEx rule generation that successfully classified **500+ log lines**.

REAL-TIME CAMERA CLASSIFIER

Designed and built a desktop computer vision application for real-time image classification using Tkinter, OpenCV, and scikit-learn.

- **Engineered** a real-time video streaming pipeline using **OpenCV** and **Tkinter**, processing webcam feeds at a **15ms refresh loop** with automated dataset capturing.
- **Implemented** a supervised machine learning classifier using **scikit-learn LinearSVC** trained on **16,950-dimensional** grayscale pixel vectors.
- **Developed** an automated prediction engine with dynamic UI state updates in **Python**, delivering live video inference and class label identification.

MICROGRAD AUTOGRAD ENGINE

Built a scalar-valued automatic gradient (autograd) engine and neural network library from scratch in Python.

- **Engineered** a scalar-valued **autograd engine** (Value class) using **topological sorting (DAG)** and calculus **chain rule** in **Python**, enabling exact reverse-mode automatic differentiation.
- **Developed** modular PyTorch-style **Neuron**, **Layer**, and **MLP** abstractions by overloading scalar operators (+, *, pow, tanh), structuring neural networks with **41+ trainable parameters**.
- **Implemented** an end-to-end training loop incorporating **MSE loss**, **gradient zeroing**, and **Gradient Descent**, driving classification loss down from **7.12 to ~0.5**.

CERTIFICATIONS

- **Machine Learning A-Z: ML, DL, AI with AWS, Python & R** -- Udemy
- **Artificial Intelligence with Python** -- Coincent.ai

HOBBIES & INTERESTS

- Chess
- Singing
- Video Games & Gameplays
- Anime